Standard Operating Procedure

SOP: Fasting Plasma Glucose and HbA1c Sample Collection and Handling

SOP ID: SPbE-2026-SOP-001 | Version 1.0 | Synthetic example.

1. Purpose

To standardize the collection, labeling, processing, and storage of venous blood samples used to measure fasting plasma glucose and HbA1c, ensuring comparable results across visits and subjects.

2. Scope

Applies to all study staff collecting biospecimens for study SPbE-2026 at baseline and week 12.

3. Materials

- Sodium fluoride / potassium oxalate tubes (grey-top) for glucose.
- EDTA tubes (lavender-top) for HbA1c.
- Pre-printed barcode labels linked to subject_id.
- Calibrated centrifuge, refrigerated transport box, validated -80 C freezer.

4. Procedure

1. Confirm the participant has fasted >= 8 hours; record last food intake time.
2. Collect blood by standard venipuncture into the grey-top tube first, then the lavender-top tube.
3. Invert each tube gently 8-10 times immediately after collection. Do not shake.
4. Label each tube with the barcode and record visit_date and time.
5. Centrifuge the glucose tube within 30 minutes (1500 x g, 10 min, 4 C); separate plasma.
6. Store HbA1c whole blood at 4 C and analyze within 72 hours; archive plasma aliquots at -80 C.
7. Log every sample in the LIMS against the subject_id before end of day.

5. Quality Control

Run two levels of commercial control material with each analytical batch. Reject and recollect any sample with visible hemolysis or an unbroken cold chain. Document deviations on the deviation log.

6. Records

Retain collection logs, instrument calibration records, and QC results. These records map directly to the dataset fields visit_date, baseline_glucose_mg_dl, week12_glucose_mg_dl, and hba1c_pct.